1. Derivatives of Activation Functions

① ReLU

$$f(x) = \begin{cases} x & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$f'(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$

* the derivative of ReLU at $x=0$ is supposed to be undefined but I will return 0

② Leaky ReLU

$$f(x) = \begin{cases} x & x \geq 0 \\ ax & x < 0 \end{cases}$$

$$f'(x) = \begin{cases} 1 & x \geq 0 \\ a & x < 0 \end{cases}$$

③ sigmoid function

$$f(x) = \frac{1}{1+e^{-x}}$$

$1+e^{-x} \Rightarrow a$
$e^{-x} \Rightarrow b$
$-x \Rightarrow c$

$$f'(x) = \frac{dy}{da} \times \frac{da}{db} \times \frac{db}{dc} \times \frac{dc}{dx}$$

$$= \frac{1}{(1+e^{-x})^2} \times 1 \times e^{-x} \times -1$$

$$= \frac{e^{-x}}{(1+e^{-x})^2}$$

$$= \frac{1+e^{-x}-1}{(1+e^{-x})^2}$$

$$= \frac{1+e^{-x}}{(1+e^{-x})^2} - \frac{1}{(1+e^{-x})^2}$$

$$= \frac{1}{1+e^{-x}} - \frac{1}{(1+e^{-x})^2}$$

$$= \frac{1}{1+e^{-x}}\left(1 - \frac{1}{1+e^{-x}}\right)$$

$$= f(x)(1-f(x))$$

④ Softmax

$$f(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}$$

Case 1. $i = j$

numerator $g(z_i) = e^{z_i}$ $\quad g'(z_i) = e^{z_i}$

denominator $h(z_i) = \sum_k e^{z_k}$ $\quad h'(z_i) = e^{z_i}$

quotient rule $\Rightarrow \frac{g'h - gh'}{h^2} = \frac{e^{z_i}\left(\sum_k e^{z_k}\right) - e^{z_i}(e^{z_i})}{\left(\sum_k e^{z_k}\right)^2}$

$$= \frac{e^{z_i}}{\sum_k e^{z_k}} - \frac{(e^{z_i})^2}{\left(\sum_k e^{z_k}\right)^2}$$

$$= S_i(1 - S_i)$$

Case 2. $i \neq j$

numerator $g(z_j) = e^{z_i}$ $\quad g'(z_j) = 0$

denominator $h(z_j) = \sum_{j=1}^{K} e^{z_j}$ $\quad h'(z_j) = e^{z_j}$

quotient rule $\Rightarrow \frac{0 \times \sum_{j=1}^{K} e^{z_j} - e^{z_i} \times e^{z_j}}{\left(\sum_{j=1}^{K} e^{z_j}\right)^2}$

$$= 0 - \frac{e^{z_i} e^{z_j}}{\left(\sum_{j=1}^{K} e^{z_j}\right)^2}$$

$$= -S_i S_j$$

$$\frac{\partial S_i}{\partial z_j} = S_i(\delta_{ij} - S_j)$$

⑤ tanh

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{\sinh(x)}{\cosh(x)}$$

$$f'(x) = \frac{\cosh(x) \times \cosh(x) - \sinh(x) \times \sinh(x)}{\cosh^2(x)}$$

$$= \boxed{1 - \tanh^2(x)}$$

2. Derivatives of Loss Functions

① MSE

$$L = \frac{1}{2n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

* used $\frac{1}{2n}$ for a more simple derivative

$$\boxed{\frac{\partial L}{\partial \hat{y}} = -\frac{1}{n}(y_i - \hat{y}_i)}$$

② MAE

$$L = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$sign = \begin{cases} -1, & y_i < \hat{y}_i \\ 1, & y_i > \hat{y}_i \\ \text{undefined} & y_i = \hat{y}_i \end{cases}$$

↳ will use zero instead

$$\boxed{\frac{\partial L}{\partial \hat{y}} = -\frac{1}{n}\,sign(y - \hat{y})}$$

③ BCE

$$L = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right]$$

$$\frac{\partial L}{\partial \hat{y}} = -\frac{1}{n}\left(\frac{y}{\hat{y}} - \frac{1-y}{1-\hat{y}}\right)$$

$$= -\frac{1}{n}\left(\frac{y - y\hat{y} - \hat{y} + y\hat{y}}{\hat{y}(1-\hat{y})}\right)$$

$$= \boxed{-\frac{1}{n}\left(\frac{y - \hat{y}}{\hat{y}(1-\hat{y})}\right)}$$

③.5 BCE with sigmoid

$$\frac{\partial L}{\partial z} = \frac{\partial y}{\partial z} \times \frac{\partial L}{\partial y}$$

$$= \hat{y}(1-\hat{y}) \times -\frac{1}{n}\left(\frac{y - \hat{y}}{\hat{y}(1-\hat{y})}\right)$$

$$= \boxed{-\frac{1}{n}(y - \hat{y})}$$

④ CCE

$$L = -\frac{1}{n}\sum_{i=1}^{n} y_i \log(\hat{y}_i)$$

$$\frac{\partial L}{\partial \hat{y}} = -\frac{1}{n}\left(\frac{y}{\hat{y}}\right)$$

④.5 CCE with softmax

$$\frac{\partial L}{\partial z_i} = \frac{\partial L}{\partial s_k} \times \frac{\partial s_k}{\partial z_i} = \sum_k -\frac{y_k}{s_k} \times s_k(\delta_{ki} - s_i)$$

$$= \sum_k -y_k(\delta_{ki} - s_i)$$

$$= \sum_k -y_k\delta_{ki} + y_k s_i$$

$$= \sum_k -y_k\delta_{ki} + \sum_k y_k s_i$$

$$= -y_i + s_i$$